

Credit Risk Modelling and Data Engineering Pipeline

An End-to-End Production Credit Risk System on the Lending Club Dataset

SOPHON Rachana LOT Soklang DUK Pagnarith LENG Devid

Supervised by **Lecturer TOEM TOUCH**

Institute of Technology of Cambodia — Department of Information Technology

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Our Story in Five Acts

- 1 Act I — The Problem
- 2 Act II — The Foundation
- 3 Act III — Building the Model
- 4 Act IV — The Reckoning
- 5 Act V — The Warning
- 6 Act VI — The Rebuild
- 7 Epilogue — What We Learned

When the Numbers Lie: The Cost of Getting Credit Risk Wrong

What Credit Risk Actually Is

"The potential that a bank borrower or counterparty will fail to meet its obligations in accordance with agreed terms."

— Basel Committee on Banking Supervision

When a borrower defaults, the lender loses:

- ▶ Outstanding **principal**
- ▶ Expected **interest income**
- ▶ **Collection and legal costs**

$$\text{Net Loss} = EAD \times LGD$$

2008: A Cautionary Tale

- ▶ PD models calibrated on **rising house prices**
- ▶ Ignored **correlated defaults** across mortgage pools
- ▶ **\$700B TARP** bailout authorised
- ▶ **+30M** global unemployed
- ▶ US unemployment peaked at **10.0%**

*The models were technically correct.
The assumptions were catastrophically wrong.*

The World's Response: Three Pillars of Regulation

Basel II/III (IRB)

Banks estimate own **PD**, **LGD**, **EAD**

Capital requirement:

$$CAR = \frac{\text{Capital}}{RWA} \geq 8\%$$

TtC PD for capital stability

IFRS 9 (ECL)

Forward-looking **Expected Credit Loss**

3-stage impairment:

$$ECL = \sum_t PD_t \cdot LGD_t \cdot EAD_t$$

PiT PD with macro scenarios

SR 11-7 (Model Risk)

Model risk management

Three pillars:

- ▶ Development & use
- ▶ Independent validation
- ▶ Governance & inventory

These frameworks transform credit scoring into a supervised, regulated modelling infrastructure.

The Gap Nobody Fills

What Prior Lending Club Studies Do

- ▶ Benchmark ML algorithms on static snapshots
- ▶ Report AUC / accuracy metrics only
- ▶ Stop at offline model evaluation
- ▶ ✗ No regulatory compliance
- ▶ ✗ No IFRS 9 / Basel III outputs
- ▶ ✗ No production monitoring

A model that scores well on a leaderboard is not the same as a model a bank can use.

Our 5 Contributions

- ➊ **Application-time features** — no target leakage
- ➋ **WoE scorecard** — Basel interpretability + SR 11-7
- ➌ **IFRS 9 & Basel III** — PiT and TtC PD alignment
- ➍ **Credit policy engine** — 10-class AA-F + ECOA
- ➎ **Production MLOps** — Airflow, MLflow, PSI/CSI

We cross that gap.

Standing on Shoulders: From Z-Scores to Gradient Boosting

Six Decades of Progress

- ▶ Beaver (1966) — univariate ratio analysis
- ▶ Altman (1968) — Z-score, MDA
- ▶ Wiginton (1980) — logistic regression
- ▶ Siddiqi (2006) — **WoE/IV scorecard**
- ▶ Thomas et al. (2002) — academic consolidation

The ML Turn

- ▶ Baesens et al. (2003) — 41-classifier benchmark
- ▶ Lessmann et al. (2015) — ensembles win on AUC
- ▶ Dong et al. (2023) — XGBoost / LightGBM
- ▶ Demajo et al. (2025) — LightGBM + SHAP

The Core Tension

Model	AUC	Regulatory
Logistic/WoE	Good	✓ Easy
Random Forest	Better	≈ Moderate
XGBoost	Best	✗ Hard

EBA (2023)

Interpretability and explainability remain the **key obstacles** to ML adoption in IRB models.

Our Answer

Logistic regression with WoE encoding — regulatory-grade interpretability, adequate discrimination (Gini 0.387).

The Scorecard Toolkit: WoE, IV, and Point Scaling

Weight of Evidence

$$WoE_i = \ln\left(\frac{\% Goods_i}{\% Bads_i}\right)$$

Positive WoE $\Rightarrow$ lower risk Negative $\Rightarrow$ higher risk

Information Value

$$IV = \sum_i (\%G_i - \%B_i) \times WoE_i$$

$IV < 0.02$	Useless
$0.02-0.10$	Weak
$0.10-0.30$	Medium
$0.30-0.50$	Strong
> 0.50	Suspicious

PDO Score Scaling

$$Factor = \frac{PDO}{\ln 2} \approx 28.85, \quad Offset = 600$$

Our Scorecard Settings

- ▶ Scale: **300–850** (FICO standard)
- ▶ PDO: **20 points** doubles odds
- ▶ Reference: **600** at 1:1 odds
- ▶ Higher score = lower PD

Missing values $\rightarrow$ **separate WoE bin.**

Missingness is behaviourally informative in credit data.

Meet the Data: 2.26 Million Borrowers, 11 Years of History

Dataset Properties

Rows	2,260,668
Columns	151
Date range	2007–2018
File size	≈1.6 GB
Default rate	≈17%
100% null	15 columns

The First Trap

File is **NOT sorted chronologically**. Must derive `issue_year` from `issue_d` for the OOT split.

57 columns cut (5 reasons):

- 1 100% null (15 cols)
- 2 Identifiers / constants (8)
- 3 Joint application > 99% null (3)
- 4 Hardship/settlement > 97% null (18)
- 5 **Post-application leakage** (15)
`total_pymnt`, `recoveries`, `out_prncp`

6 derived features added:

`fico_score`, `term_int`, `int_rate` ($\div 100$),
`mths_since_issue_d`,
`mths_since_earliest_cr_line`, `emp_length_int`

Rule Zero: Never Train on What You Cannot See at Origination

Three target variables:

$$good_bad = \begin{cases} 0 & \text{Charged Off / Default} \\ 1 & \text{Fully Paid} \end{cases}$$

$$recovery_rate = \frac{recoveries}{funded_amnt}$$

$$ccf = \frac{funded_amnt - total_pymnt}{funded_amnt}$$

Out-of-Time Split

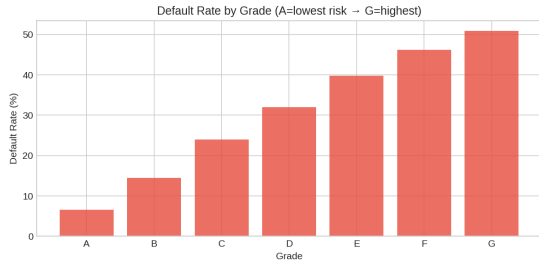
Set	Years	Rows	DR
Train	2007–15	831K	18.6%
OOT	2016–18	539K	25.3%

Why Not Random Split?

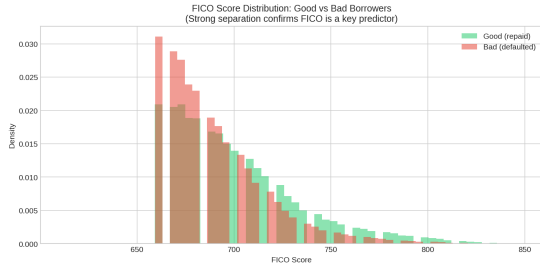
Random split leaks future patterns into training. OOT mirrors real deployment: train on history, score the future.

Reject inference (parceling): 26,129 synthetic rejected applicants added to training set to partially correct selection bias.

What the Data Tells Us About Risk

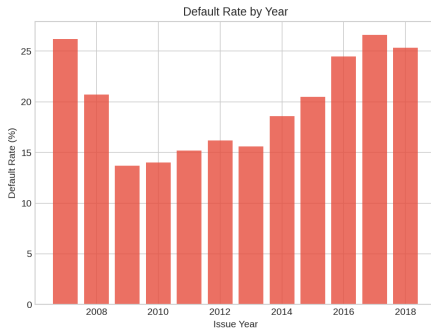
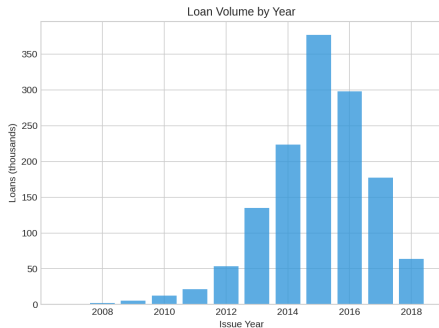


Default rate **monotonically rises** Grade A ($\approx 5\%$) → G ($> 35\%$). Validates WoE ordering and 10-class credit policy.



FICO separates good and bad borrowers. $IV > 0.30$ confirms it is the **strongest** model predictor.

A Storm Building Since 2015



Training Window 2007–2015

Default rate $\approx 18.6\%$ — stable. WoE bins fitted on this period.

OOT 2016–2018: The First Warning

Default rate jumps to **25.3%** (+6.7pp). First signal of population drift, confirmed later by **Score PSI = 0.282**.

A Judge That Can Be Cross-Examined

Logistic Regression (statsmodels)

$$\log \frac{P(\text{Good})}{P(\text{Bad})} = \beta_0 + \sum_{j=1}^{39} \beta_j \cdot \text{WoE}_j$$

Feature selection: iterative backward elimination
— remove variable with highest max p-value until all $p < 0.05$ (Wald test).

Candidate WoE dummies	56
Retained (significant)	39
Raw predictors	16

Why statsmodels? Proper **Wald-test p-values** and confidence intervals required by EBA/GL/2017/16.

PDO Scaling

$$s_j = -(\hat{\beta}_j \times 28.85); \text{sum} + \text{intercept offset}$$

Calibration ($\delta = +0.3204$)

$$\hat{P}^{calib} = \sigma(\text{logit}(\hat{P}^{raw}) + \delta)$$

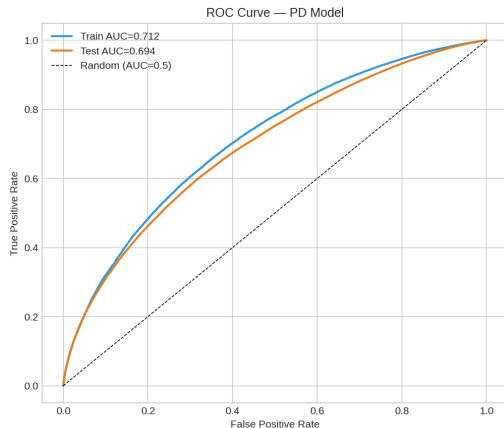
	Pre	Post
Mean PD	20.2%	25.3%
HL stat	10,039	403 (−96%)

Two PD Variants

PiT PD → IFRS 9 ECL

TtC PD → Basel AIRB capital

Does Our Model Actually Work?



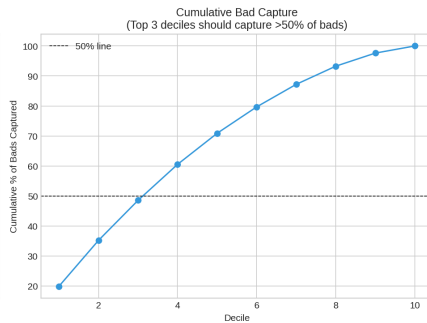
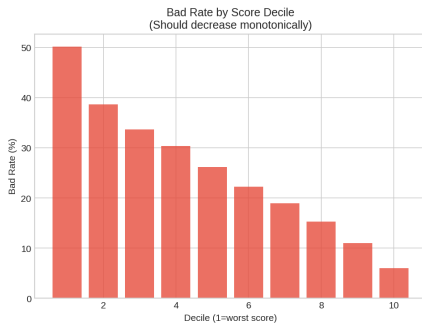
Metric	Value	Bench.	
AUC	0.694	> 0.65	✓
Gini	0.387	≥ 0.40	≈
KS	0.281	> 0.25	✓
Brier	0.172	≤ 0.20	✓
Mean PD	25.3%	Matches DR	✓
Score PSI	0.282	< 0.25	✗

Score PSI = 0.282 (ALERT)

Population shifted. Model **still discriminates** (Gini 0.387 maintained) but requires rebuild on recent data.

AUC = 0.694, Gini = 0.387. Curve lies above the diagonal — genuine discriminatory power confirmed.

The Ranking Test: Does Higher Score Mean Lower Risk?



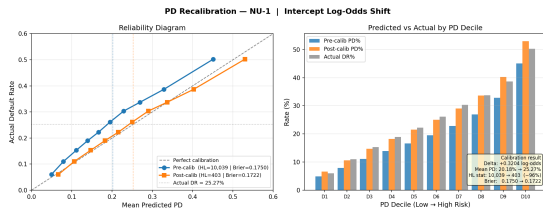
Monotonic Ordering

Bad rate decreases **strictly** Decile 1→10 — **zero violations**. Mandatory acceptance criterion (Siddiqi, 2006).

Bad Borrower Concentration

Deciles 1–3 capture > 50% of bad borrowers (only 30% of portfolio). Directly validates the auto-reject rule for the F class.

Plot Twist: The Model Was Systematically Underpredicting



Before: mean PD = 20.2% vs actual 25.3%. All decile points fall **below** the 45° diagonal — systematic under-prediction.

After: $\delta = +0.3204$ brings mean PD to 25.3%. Points align with the diagonal.

The Financial Stakes

Without calibration:

EL under-estimated by **+\$158M (24%)**

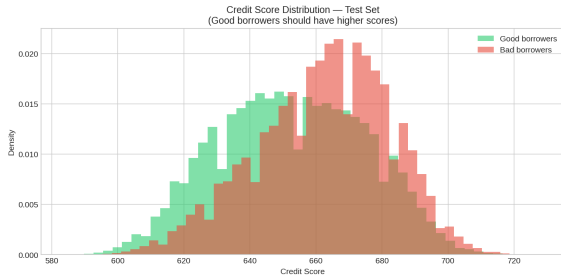
ECL under-estimated by **+\$267M (20%)**

Stage 2 pool too small by **13.2pp**

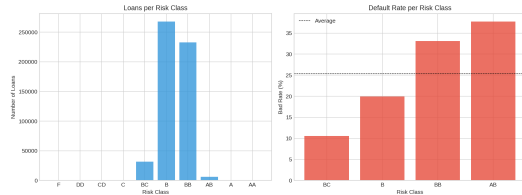
Calibration is not cosmetic.

It cascades through every regulatory output.

Seeing the Borrower Population Through Its Scores



Good borrowers cluster at **higher scores**; bad at **lower scores**. The 500–650 overlap is where the ROI-based policy adds value.



Portfolio concentrates in **BB–C** bands.

Auto-approve (AA/A) and auto-reject (F) cover the tail extremes. Most decisions use the ROI rule.

Who Gets the Loan? The 10-Class Policy Engine

Class	Score	Decision
AA	780–850	Auto-Approve
A	740–779	Auto-Approve
AB–CD	700–539	Approve if $ROI > r_f$
DD	460–499	Manual Review
F	300–459	Auto-Reject

ROI formula:

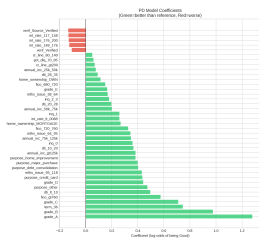
$$ROI_{ann} = \frac{\text{interest income} - EL}{\text{funded_amnt} \times \frac{\text{term}}{12}}$$

$r_f = 2.15\%$ (US base rate at time of data)

ECOA Compliance

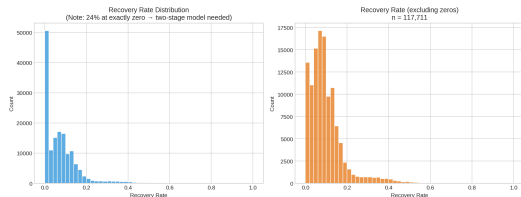
Every rejection generates up to **4 adverse action codes** derived from scorecard contributions.

e.g. “AA02 — Debt-to-income ratio too high”



All 39 WoE dummies significant ($p < 0.05$). Positive = lower default risk; negative = higher risk.

When Things Go Wrong: How Much Do We Actually Lose?



$\approx 50\%$ zero recovery justifies two-stage design.

Stage 1 — Logistic: $P(RR > 0)$

Stage 2 — Linear: $E[RR \mid RR > 0]$

Combined: $L\hat{G}D = 1 - \hat{P}_1 \times \hat{R}R_2$

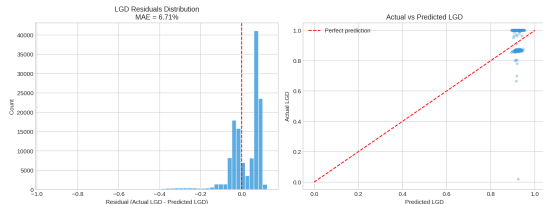
LGD MAE	$\approx 5\%$
Mean LGD (LR)	92.19%
Downturn LGD	93.76%

Why LGD Is So High

Unsecured personal loans $\Rightarrow$ **no collateral**.

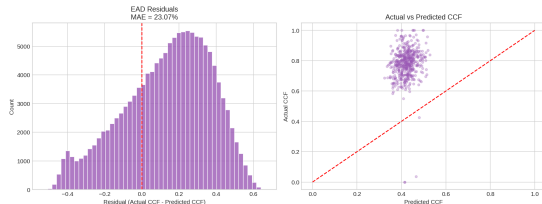
Average recovery only $\approx 8\%$. This drives the high EL rate (19.93%) and capital ratio (34.5% of EAD).

Checking Our Work: Model Residuals



LGD MAE $\approx$ 5%

Centred around zero. Slight negative skew:
conservative underestimate of large recoveries.



EAD MAE $\approx$ 14%

Larger spread than LGD — predicting *when* default
occurs within the loan lifecycle is inherently
uncertain. No directional bias.

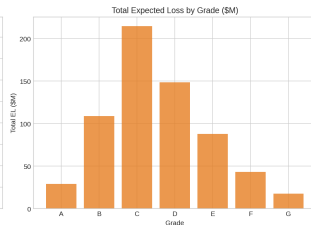
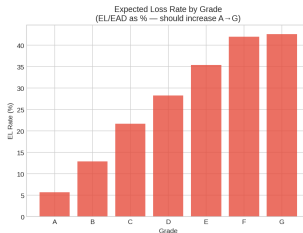
Counting the Cost: \$808 Million in Expected Losses

$$EL_i = \hat{PD}_i^{PiT} \times \hat{LGD}_i \times E\hat{AD}_i$$

Total EAD	\$3.26B
Mean LGD	92.19%
12-month EL	\$0.808B
EL rate	19.93%

Key v4 Correction

Prior versions used **grade-proxy PD**. Now uses the **real calibrated PiT PD** from the scorecard — a 24% increase in computed EL.



EL rate rises **monotonically** A → G. Steeper than the default rate gradient because lower-grade borrowers also have higher EAD.

Sorting Borrowers by Their Future: IFRS 9 Staging

SICR Classification:

- **Stage 3** (credit-impaired): 90+ DPD or Charged Off
- **Stage 2** (SICR): $PD \geq 2 \times \text{origin PD}$ or $\Delta PD \geq 5pp$ or 30-DPD backstop
- **Stage 1** (performing): none of above

ECL formulas:

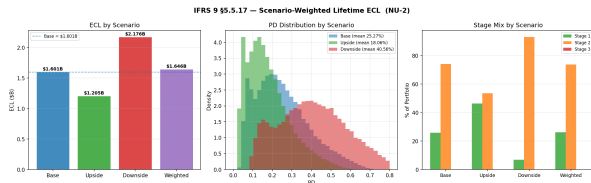
$$ECL^{12m} = PD \cdot LGD \cdot EAD \cdot \min\left(\frac{term}{12}, 1\right)$$

$$ECL^{life} = \sum_{t=1}^T \frac{h(1-h)^{t-1} LGD \cdot EAD}{(1+r)^t}$$



Stage 2 dominates at 71.4% (\$967M). The calibrated PiT PD of 25.3% causes many OOT loans to trigger the “PD doubling” SICR rule.

Preparing for the Worst: Probability-Weighted Scenario ECL (§5.5.17)



Macro satellite model:

$$\Delta \logit(PD) = 0.18\Delta u - 0.10\Delta g$$

Scen.	Wt	PD	ECL
Base	50%	25.3%	\$1.60B
Upside	25%	18.1%	\$1.21B
Downside	25%	40.6%	\$2.18B
Wgtd	—	—	\$1.65B

Non-linearity Uplift

Weighted ECL exceeds base by **+\$44M**.
Downside is +36% above base. Lifetime ECL is **convex in PD**: losses accelerate faster than PD rises.

How Much Safety Net Does a Bank Need? Basel III Capital

Retail IRB formula:

$$\rho = 0.03 + 0.16 \cdot \frac{1 - e^{-35PD}}{1 - e^{-35}}$$

$$PD_{WC} = \Phi\left(\frac{\Phi^{-1}(PD) + \sqrt{\rho}\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right)$$

$$K = LGD_{DT}(PD_{WC} - PD), \quad RWA = 12.5 K EAD$$

$$LGD_{DT} = 93.76\% \text{ (downturn; } 1.25\times \text{ haircut, Basel CRE36)}$$

The Answer: \$1.13B

Total EAD	\$3.26B
IRB RWA	\$14.07B
Min Capital (8%)	\$1.13B
Capital / EAD	34.5%

Why the Ratio Is Unusually High

Capital scales with **LGD through the AIRB formula**. At $LGD \approx 93\%$, unexpected losses are large. Secured portfolios ($LGD \approx 40\text{--}60\%$) carry far lower capital ratios.

The Full Picture: From a Single Loan to a \$3.26B Portfolio

Complete Risk Stack

Model outputs

Total EAD	\$3.26B
Mean LGD (LR)	92.19%
Downturn LGD	93.76%

Expected loss

12-month EL	\$0.808B
EL rate (% EAD)	19.93%

IFRS 9 lifetime ECL

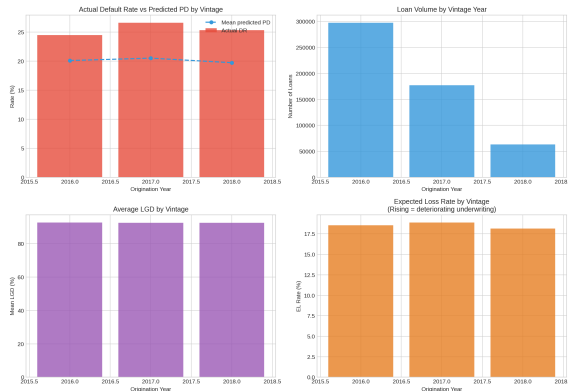
Base scenario	\$1.601B
Upside	\$1.205B
Downside	\$2.176B

Weighted	\$1.646B
vs 12m EL uplift	+\$838M (+104%)

Basel III capital

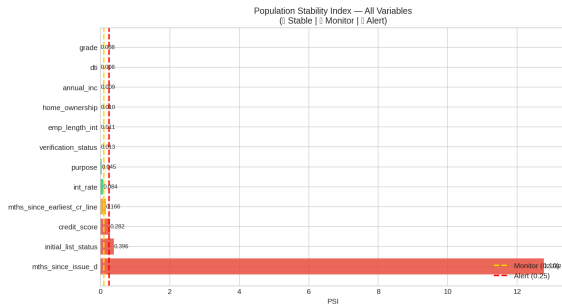
IRB RWA	\$14.07B
Min Capital	\$1.13B

Vintage Cohort Analysis — OOT Test Portfolio (2016–2018)



Predicted PD tracks actual default rate across vintage cohorts. **2016+ cohorts** show widening gap — the quantitative signal for a model rebuild.

The Alert We Hoped Not to See: $PSI = 0.282$



Score PSI bar is in the **red ALERT zone** at 0.282.
Several CSI bars also elevated — root-cause signals for which features drive the drift.

PSI thresholds:

PSI	Status	Action
< 0.10	Stable	None
0.10–0.25	Monitor	Investigate
> 0.25	Alert	Rebuild

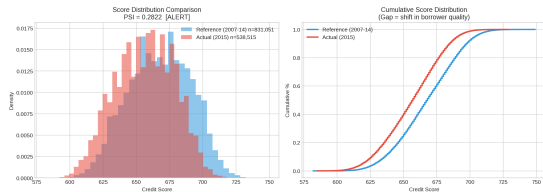
$$PSI = \sum_i (O_i - E_i) \ln(O_i / E_i)$$

O_i = OOT bin proportion, E_i = Training bin proportion

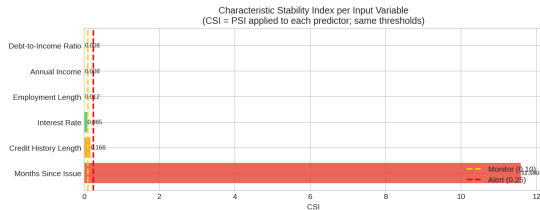
Score PSI = 0.282

ALERT. Model discriminates (Gini 0.387 maintained) but the applicant population has changed materially — retrain on 2016–2018 data is recommended.

Who Changed? Diagnosing the Population Shift

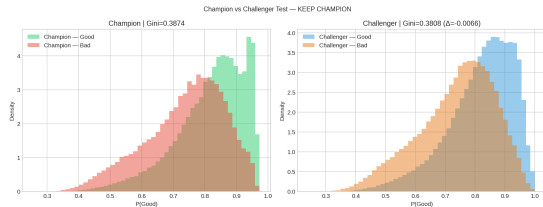


OOT population (2016–2018) shifted to lower scores vs training. Consistent with 25.3% OOT default rate vs 18.6% training rate.



CSI by variable: high-IV predictors (greatest scorecard contribution) are also the most unstable — flagged for re-binning in the next model build.

Is There a Better Model? Challenging the Champion



Reject Inference (Parceling):

	Original (831K)	Augmented (857K)
Gini	0.3874	0.3879
AUC	0.6937	0.6940
Mean PD	20.2%	20.7%

Verdict: Keep Champion

Challenger Gini ≈ 0.387 — identical.
Mann-Whitney U: not significant ($< 2pp$ threshold). **Next challenger:**
XGBoost/LightGBM.

Interpretation

Discrimination: unchanged. Meaningful effect on **PD level**: +0.52pp correction toward the true population default rate.

The Alert Triggered a Rebuild: Strategy

Trigger: Score PSI = 0.282

Champion (2007–2015) operates outside its training domain. Rebuild required.

Rebuild Strategy:

- ▶ **Retain** WoE bin structure from L01
- ▶ **Re-estimate** logistic $\hat{\beta}$ on 2016–2017 data
- ▶ **Calibrate** to 2018 observed DR (25.33%)
- ▶ **Evaluate** on 2018 holdout vs Champion

Set	Years	Rows	DR
Champ train	2007–15	831K	18.6%
Rebuild train	2016–17	475K	25.3%
Holdout	2018	64K	25.3%

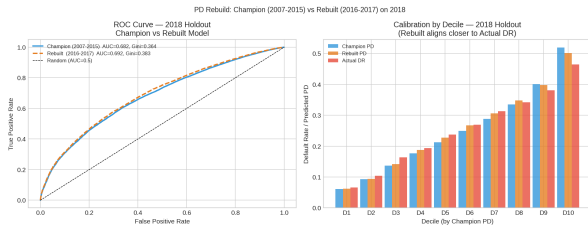
What Changes vs What Stays

Component	Champ	Rebuilt
WoE bins	2007–15	2007–15
$\hat{\beta}$	2007–15	2016–17
δ	+0.320	+0.067
Features	56	39

Why Same WoE Bins?

Re-binning requires the full raw pipeline. Re-estimating logistic weights is the “fast-track” rebuild — corrects level bias, updates risk weights, no architecture change.

The Rebuild Works: +1.9pp Gini, Calibration Corrected



Rebuild ROC (orange dashed) lies above Champion (blue).
Right: decile calibration — Rebuild aligns to actual DR (red bars).

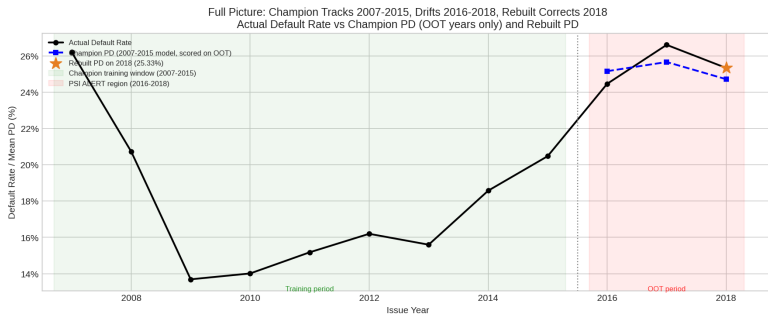
Metric	Champ	Rebuild	Δ
AUC	0.682	0.692	+0.010
Gini	0.364	0.383	+0.019
KS	0.268	0.280	+0.012
Brier	0.175	0.173	-0.002
Mean PD	24.7%	25.3%	+0.6pp

Verdict: Strong Challenger

Gini +1.9pp — just below 2pp promotion threshold.
All metrics improved. Full WoE re-binning is the next step before formal promotion.

Champion -0.62pp underprediction eliminated.
PSI(Champ vs Rebuild on 2018) = 0.009 (STABLE).

The Full Story: Champion Drift and Rebuilt Correction



2007–2015: Champion Tracks Well

Trained on this window — predicted PD follows observed DR closely.

2016–2018: Drift and Correction

Champion underestimates. Rebuilt (★) anchors at 25.3% — the correct 2018 level.

What the Journey Revealed: Six Findings

1. Calibration is financially material

Without $\delta = +0.3204$: EL under-stated by **\$158M (24%)**, ECL by **\$267M (20%)**, Stage 2 pool by **13.2pp**.

2. Lifetime ECL $\gg$ 12-month EL

\$1.646B vs \$0.808B — uplift **+\$838M (+104%)**.
Stage 2 loans (74.2%) carry lifetime ECL under IFRS 9.

3. LGD drives everything

LGD 92.19% $\Rightarrow$ capital ratio 34.5% of EAD.
Unsecured portfolios require far more capital than secured lending.

4. Discrimination adequate, calibration needed

Gini 0.387, KS 0.281 meet benchmarks.
Post-calibration HL statistic drops 96%.

5. The population has shifted

Score PSI 0.282 (ALERT). Model ranks borrowers correctly but operates outside its training domain.
Rebuild required.

6. The production gap is vast

Reject inference, OOT splitting, PiT/TtC calibration, IFRS 9 staging, Basel capital, ECOA codes, PSI monitoring — this is the gap this project crosses.

Being Honest: What We Got Wrong and What We Couldn't Do

Model Limitations

- ❶ **RAROC +299% inflated:** computed on full portfolio. Meaningful RAROC requires approved-only subset.
- ❷ **Score PSI = 0.282 (ALERT):** model needs retrain on 2016–2018 cohort.
- ❸ **Reject inference limited:** only 3.1% synthetic rows; full correction requires real rejected-loan data.

Regulatory Limitations

- ❹ **Macro sensitivities literature-based:** β_{unem} , β_{GDP} need empirical estimation from this dataset's time series.
- ❺ **No external validation:** OOT holdout only. SR 11-7 §4 independent review not yet performed.
- ❻ **P2P context:** LGD 93% is specific to unsecured consumer lending — not transferable to secured portfolios.

What Comes Next

Short-term (Next Iteration)

- ① ✓ **PD rebuild** on 2016–2017 data — completed (Act VI). Full WoE re-binning remains.
- ② **Gradient boosting challenger** (XGBoost / LightGBM) with SHAP for adverse action code generation
- ③ **Empirical macro sensitivities** from dataset time-series

Medium-term (Production)

- ④ **Independent model validation** (SR 11-7 §4) before any regulatory use
- ⑤ **RAROC on approved-only subset** for meaningful risk-adjusted return
- ⑥ **Airflow DAG productionisation** and Grafana dashboard completion

The Journey Doesn't End Here — But We've Crossed the Gap

What We Built

An end-to-end, reproducible credit risk platform on 2.26M Lending Club loans integrating:

- ▶ WoE logistic scorecard with SR 11-7 governance
- ▶ IFRS 9 three-stage ECL with scenario weighting
- ▶ Basel III AIRB capital calculation
- ▶ Daily PSI/CSI monitoring with alerting
- ▶ 10-class credit policy + ECOA adverse action codes

Verified

All 4 notebooks ran end-to-end on the real dataset with **zero errors** (v4 run, 2026-06-07).

The Numbers:

Gini (OOT)	0.387
AUC (OOT)	0.694
Score PSI	0.282 ALERT
12m EL	\$0.808B
IFRS 9 ECL	\$1.646B
RWA	\$14.07B
Capital	\$1.13B

“The gap between a trained model and a production-grade, regulatory-compliant credit risk system is large. This project maps and crosses that gap.”

Thank You

Questions & Discussion

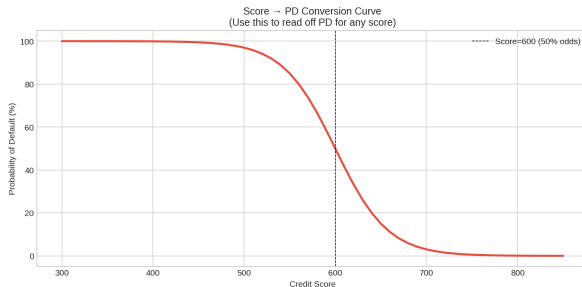
SOPHON Rachana · LOT Soklang · DUK Pagnarith · LENG Devid

Supervised by **Lecturer TOEM TOUCH**

Institute of Technology of Cambodia

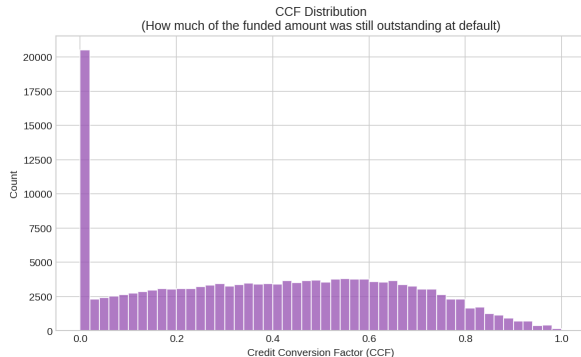
Academic Year 2025–2026

Backup: Score-to-PD Conversion



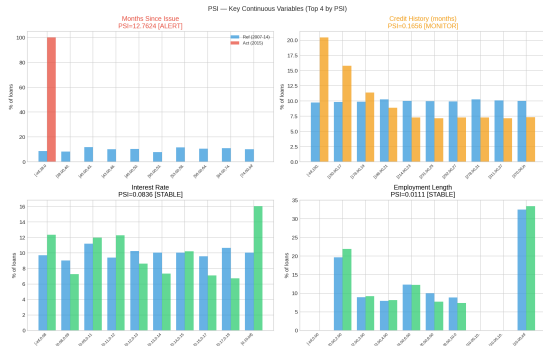
The score-to-PD mapping is monotonically decreasing. Every 20-point increase halves default odds (PDO = 20). Reference: score 600 at odds 1:1. This mapping is the basis for IFRS 9 SICR classification and ECOA adverse action documentation.

Backup: CCF Distribution

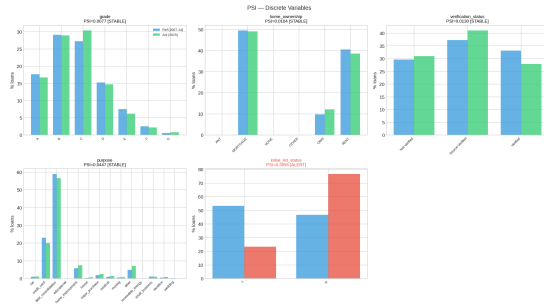


CCF is right-skewed with mass near 1.0. Most defaults occur early before significant repayment. Median CCF > 0.8 means EAD typically exceeds 80% of funded amount.

Backup: PSI by Variable Type



Continuous CSI: red bars (> 0.25) are primary drivers of score shift.



Discrete CSI: categorical mix shift in the 2016–2018 OOT population.

Backup: PD Calibration Impact

Output	Pre-calib.	Post-calib.	Δ
Mean PD	20.18%	25.27%	+5.09pp
H-L statistic	10,039	403	-96%
IRB RWA	\$13.62B	\$14.07B	+\$0.45B
IRB Min Capital	\$1.09B	\$1.13B	+\$0.04B
Capital / EAD	33.4%	34.5%	+1.1pp
12-month EL	\$0.650B	\$0.808B	+\$0.158B (+24%)
IFRS 9 lifetime ECL	\$1.334B	\$1.601B	+\$0.267B (+20%)
IFRS 9 Stage 2 mix	61.0%	74.2%	+13.2pp

Calibration is not cosmetic — it cascades through every regulatory output.

Backup: Reject Inference Detail

Metric	Original (831K)	Augmented (857K)	Δ
AUC	0.6937	0.6940	+0.0003
Gini	0.3874	0.3879	+0.05pp
KS	0.2813	0.2816	+0.0003
Brier Score	0.1750	0.1751	+0.0001
Mean OOT PD	20.18%	20.70%	+0.52pp
Train DR	18.62%	20.35%	+1.73pp

Discrimination unchanged (3.1% synthetic rows). Meaningful effect on PD level only.